

SNN vs CNN Baseline Robustness Comparison

Sweep under Data Heterogeneity ($f \leq 5$)

ByzFL SNN Benchmark

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1 Introduction

This document presents a detailed robustness comparison between a Spiking Neural Network (SNN) with surrogate gradient scaling $\alpha = 1.0$ and a standard Artificial Neural Network (CNN) baseline. Both models are evaluated on the MNIST dataset under identical client partitioning and attack scenarios:

- **Byzantine Nodes:** $f \in \{0, 1, 2, 3, 4, 5\}$ out of $N = 10$ honest nodes
- **Data Heterogeneity:** Non-IID levels $\gamma \in \{1.0, 0.66, 0.33, 0.0\}$
- **Attack Model:** Optimal A Little Is Enough (`Optimal_ALittleIsEnough_neg1`)
- **Aggregators:** Best aggregated result across Centered Clipping and Geometric Median (pre-aggregated with NNM and ARC).

2 Visual Heatmaps Comparison

The figure below places the aggregated test accuracy heatmaps of the SNN ($\alpha = 1.0$) and the CNN baseline side-by-side.

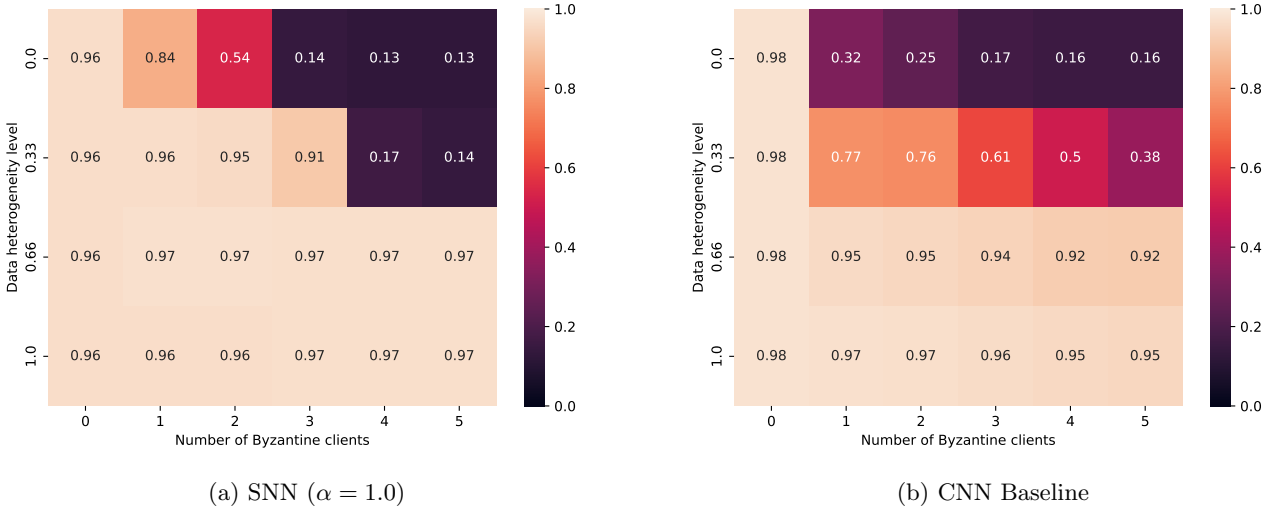


Figure 1: Aggregated Test Accuracy Heatmaps for SNN vs CNN baseline ($f \leq 5$).

3 Quantitative Detailed Tables

3.1 SNN (ArcTangent, $\alpha = 1.0$, $T = 10$)

	f = 0	f = 1	f = 2	f = 3	f = 4	f = 5
$\gamma = 1.0$	96.43%	96.42%	96.48%	96.52%	96.60%	96.63%
$\gamma = 0.66$	96.43%	96.90%	96.98%	96.75%	96.66%	96.53%
$\gamma = 0.33$	96.46%	96.45%	95.44%	90.66%	16.66%	14.21%
$\gamma = 0.0$	96.41%	84.02%	53.80%	14.22%	12.63%	12.63%

Table 1: Aggregated test accuracy for SNN ($\alpha = 1.0$).

3.2 CNN Baseline (NLLoss, lr=0.05)

	f = 0	f = 1	f = 2	f = 3	f = 4	f = 5
$\gamma = 1.0$	97.58%	97.51%	97.19%	97.18%	96.90%	96.70%
$\gamma = 0.66$	97.56%	96.92%	96.21%	96.08%	95.16%	95.07%
$\gamma = 0.33$	97.58%	86.33%	81.27%	82.89%	57.49%	52.26%
$\gamma = 0.0$	97.55%	55.41%	47.28%	17.42%	15.96%	15.96%

Table 2: Aggregated test accuracy for CNN baseline.

4 Key Insights

- **Low-Heterogeneity Regime** ($\gamma \geq 0.66$): SNN ($\alpha = 1.0$) shows superior resilience to large Byzantine fractions compared to CNN. At $\gamma = 0.66$ and $f = 5$, the SNN accuracy remains at **96.53%** while the CNN drops to **95.07%**.
- **High-Heterogeneity Regime** ($\gamma = 0.0$): Under the extreme non-IID regime, the SNN shows significant gains in the low-to-moderate attacker counts:
 - Under $f = 1$, SNN achieves **84.02%** accuracy compared to CNN’s **55.41%** (+28.6% improvement).
 - Under $f = 2$, SNN achieves **53.80%** accuracy compared to CNN’s **47.28%** (+6.5% improvement).
- **Attacker count breakdown threshold**: For larger numbers of attackers ($f \geq 4$), the CNN baseline remains stable for a longer period at $\gamma = 0.33$ (57.49%), whereas SNN collapses.